# Instructor Solution

Solution

CADET _____ SECTION _____TIME OF DEPARTURE _____

DEPARTMENT OF CHEMISTRY

QUIZ 5 – CH402 2025-2026
10 Minutes, 24 Points
6 March 2026

TEXT: McCabe, Smith, and West
SCOPE: Engineering Economics

References Permitted: FE Reference Handbook

**INSTRUCTIONS**

1. Do not mark this quiz until "begin work" is given. You will have 10 minutes.
2. Solve the problems in the space provided. Show all work to receive credit.
3. There are 5 problems on 3 pages in this quiz, not including the cover page.
4. Write your name on the top of each sheet.
5. Show work to receive partial credit.

(TOTAL WEIGHT: 24 POINTS)

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DO NOT WRITE IN THIS SPACE

| PROBLEM | VALUE | CUT |
|---|---|---|
| 1 | 5 | d |
| 2 | 4 | c |
| 3 | 5 | a |
| 4 | 5 | c |
| 5 | 5 | a |
| CUT | | |
| GRADE | 24 | |

Solution

1. (5 points) What total amount of funds will be available 12 months from now after 21% taxes if $5,000 is invested at an annual interest rate of 6% compounded monthly?

(A) $5,308
(B) $7,430
(C) $4,709
(D) $5,244

$$\$5{,}000\left(1+\frac{.06}{12}\right)^{12} = \$5{,}308$$

$$(\$5{,}308 - \$5{,}000)(1-.21) = \$244$$

$$\$5{,}000 + \$244 = \underline{\$5{,}244}$$

ANS D

2. (4 points) A proposed chemical plant will require a working-capital investment of $5 million. It is estimated that the working capital will be 21.0 percent of the total capital investment. If annual depreciation costs are estimated to be 10 percent of the fixed-capital investment, the annual depreciation (in millions of dollars) is most nearly

A) 1.3
B) 1.5
C) 1.9
D) 2.1

$$5 = .21\ TCI \Rightarrow TCI = 23.8$$

$$FCI = 23.8 - WC = 23.8 - 5 = 18.81$$

$$d = 18.81 \cdot 0.1 = \underline{1.88} \text{ million dollars}$$

ANS C

Solution

3. (5 points) Two pumps are being considered for pumping water from a reservoir. Installed cost and salvage value for the two pumps are given below:

| | Pump A | Pump B |
|---|---|---|
| Installed cost | $15,250 | $29,950 |
| Salvage value | $3,300 | $1,310 |

If the service life of Pump B is 5 years and the effective annual interest rate is 4% for both pumps, what is the service life of Pump A at which the two pumps are competitive?

A) 2.0 years
B) 3.0 years
C) 4.0 years
D) 5.0 years

$(A/F, 4\%, 5) = .1846$

$$29950 + \frac{(29950 - 1310) \cdot .1846}{.04} = 162124$$

$$15250 + \frac{(15250 - 3300) \cdot X}{.04} = 162124$$

$$x = .4916$$

use tables to match $(A/F, 4\%, n) = .4916$ gives $n = 2$

ANS a

4. (5 points) The fixed capital investment for an existing chemical plant is $20 million. Annual property taxes amount to 1% of the fixed-capital investment, and state income taxes are 5% of the gross earnings. The net income after all taxes is $2 million, and the federal income taxes amount to 25.7% of gross earnings. What are the gross earning for this plant?

A) $1.50 million
B) $2.32 million
C) $3.17 million
D) $1.66 million

$$X - .257X - .01 \cdot 20 - .05X = 2$$

$$X = 3.1746 \text{ million \$}$$

ANS C

5. (5 points) The following series of cash flows are projected for a proposed project:

| Year End | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Cash Flow | -$3,135 | $1,000 | $1,100 | $1,200 | $1,300 |

At 6% interest per year compounded annually, the *present* value of the project without taxes is most nearly

A) $825
B) $1,465
C) $3,900
D) $7,095

$(P/A, 6\%, 4) = 3.4651$

$(P/G, 6\%, 4) = 4.9455$

$-3135 + 1000 \cdot 3.4651 + 100 \cdot 4.9455$

$= \$824.65$

ANS